

| STAT214 Chapter 1 An Introduction to Veridical Data Science

| 1 The Role of Data and Algorithms in Real-World Decision Making

| 1.1 The Role of Algorithm

- Obtaining data-driven insights typically involves implementing algorithms that are designed to summarize and extract the relevant patterns and relationships that are hidden within a dataset.
- An algorithm is a set of instructions that tell your computer how to perform a specific task (such as summarizing a particular pattern).

| 1.2 The Role of Data

The relevance of an algorithm to the real world is upper-bounded by the extent to which the data that underlies it reflects reality.

| 1.3 Data is Almost Never Perfect

Data almost never perfectly captures the real world that it was designed to measure:

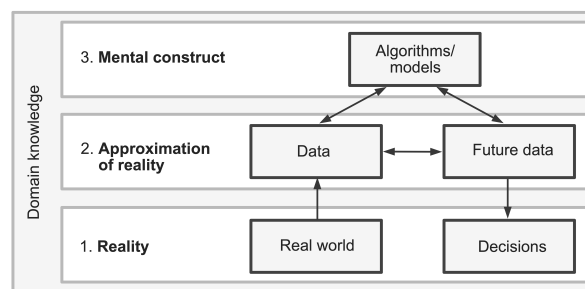
- poorly thought-out or poorly followed data collection protocols
- we typically don't know how accurate the values in any given dataset are
- mistaken assumptions may unintentionally diminish the relationship between our data and reality

These issues are not only relevant for the **current data** that we use to conduct our analyses and fit our algorithms, they are also true for the **future data** to which we will apply them.

| 1.4 How to Make Trustworthy Decisions

We need strong connections between the **three realms**:

- the real-world realm
- the data realm
- the algorithmic realm



🔗 Logic ▾

Since your data will not inform you whether it is accurately representing reality, nor will your algorithms warn you when they are being used inappropriately, you need to conduct some **data-detective** (critical thinking) work to determine whether your data is a reasonable reflection of reality and whether your algorithms are being applied appropriately to your data and domain problem.

| 2 Evaluating and Building Trustworthiness Using Critical Thinking

| 2.1 Some Critical Thinking Techniques

- Asking questions about the *domain problem*.
- Asking questions about the *data*.
- Asking questions about the *analysis, algorithms, and results*.
- Keep an eye out for *confirmation bias* and *desirability bias*.

2.2 Asking Questions about the Data

In the cases study: Estimating Animal Deaths Caused by the 2019 Australian Bushfires:

- A lot can change in 13 years
- It is a big claim to say that the estimates collected from a small, localized habitat can be generalized to the entire region affected by the fires

Therefore, the data that underlie this study is thus relevant, but there is no doubt that it is limited in scope.

⚠ Remark: Suggestion ▾

Often, **the best data is just the data you have**.

2.3 Asking Questions about the Analysis

In the cases study: Estimating Animal Deaths Caused by the 2019 Australian Bushfires:

- Does not take into account differences in animal densities across the entire affected area
- Does not take into account how these densities may have changed over time.

⚠ Remark: Suggestion ▾

Always **be transparent about the assumptions that underlie your data and results** and document them (publicly, if possible).

💡 Logic ▾

While these critical thinking questions help provide a general *qualitative* measure of the trustworthiness of data-driven results, it is often important to also provide *quantitative* measures. This is where the predictability, computability, and stability (PCS) principles come in.

3 Evaluating and Building Trustworthiness Using the PCS Framework

3.1 The PCS Framework

- **Predictability**: Your conclusions are considered to be predictable if you can show that they reemerge in relevant future data and/or are in line with existing domain knowledge.
- **Computability**: You typically don't need to demonstrate that your results are computable; rather you will use computation to conduct and evaluate your analyses, as well as to create data-inspired simulations.
- **Stability**: Your conclusions are considered stable if you can show that they remain fairly consistent when realistic perturbations (changes/modifications) are made to the underlying data, judgment calls, and algorithmic choices.

3.2 Predictability

- The most powerful predictability evaluation technique involves demonstrating that your results reemerge in the context of the actual external or future data to which you will be applying them.
- When you don't have access to any external or future data, you can emulate future or external data by withholding some of your original data *before conducting your analysis*.

3.2.1 The Training, Validation, and Test Datasets

- **training set**: at least 50 percent of the original data; used to conduct your explorations, analyses, and to train your algorithms
- **validation set**: typically use 20 percent to 40 percent of the original data; used to demonstrate that your results reemerge
- **test set**: typically 20 percent of the original data; used to provide a final independent assessment of an algorithm

3.2.2 Dataset Splitting

- **Time-based split**: If your data was collected over time and you will be applying an algorithm that you have trained to future data
- **Group-based split**: If your data has a natural grouping, and you will be applying an algorithm that you train to a new group of data points
- **Random split**: If the current data points are all more or less exchangeable with one another

⚠ Remark: Suggestions ▾

- Splitting your data in the very early stages of your project will help prevent **data leakage**
- There is no specific need to feel tied to the 60/40 train/validation split (or 60/20/20 train/validation/test split)

3.2.3 The Role of Domain Evidence in Demonstrating Predictability

another technique that we will sometimes use to demonstrate the predictability of our results is to show that our analysis or algorithm **uncovers well-established patterns or relationships in the domain field**

3.3 Stability

3.3.1 Three Sources of Uncertainty

- the data collection process using appropriate data perturbations
- the data cleaning and preprocessing judgment calls
- the choice of algorithm used

3.3.2 Investigate the Uncertainty

- recompute the results using slightly different perturbed versions of the data
- compare the results that are computed using alternative versions of the data that have been cleaned and preprocessed
- evaluate the results using future data